

UNIT – 5
ASSESSMENT – 1
PROBLEM

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Title 1: Hybrid AI Recommendation Engine Using PySpark RDDs

Problem Statement

Design a hybrid recommendation system that combines:

- **Content-Based Filtering**
- **Collaborative Filtering**
- **PySpark RDDs**
- **Intelligent Agent**

The system should analyze user-item interactions and item characteristics, generate personalized recommendations, and adapt recommendations based on user feedback.

The system should demonstrate how RDD transformations can process user-item data in a distributed manner.

Title 2: Smart Disaster Detection Using PySpark and Intelligent Agents

Problem Statement

Design a distributed AI system capable of detecting disasters such as floods and earthquakes using information from:

- Sensors
- Satellite observations
- Social media

The system must combine information from multiple sources and generate an early warning when sufficient evidence of a disaster is detected.

Title 3: AI-Based Fraud Detection Framework Using PySpark

Problem Statement

Design a scalable fraud detection framework that processes a large number of financial transactions using PySpark RDDs.

The system should:

- Identify unusual transactions.
- Detect suspicious patterns.
- Generate alerts.
- Use intelligent agents to improve fraud detection.

Title 4: Autonomous Supply Chain Optimization Using PySpark

Problem Statement

Design a distributed AI system for optimizing:

- Inventory
- Logistics
- Demand
- Transportation cost

The system should use PySpark RDDs to process supply-chain data and intelligent agents to make decentralized decisions.

Title 5: Personalized Learning Analytics Platform Using PySpark

Problem Statement

Design an AI-based educational platform that analyzes student learning behavior and performance using PySpark RDDs.

The system should:

- Analyze student scores.
- Identify weak subjects.

- Calculate learning performance.
- Recommend suitable learning content.
- Adapt recommendations based on student feedback.